

Chaos and Invariant Measures in the Logistic Map: from Fixed Points to a Markov Coarse-Graining

Arjun Sharma

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Abstract

This is a study of the logistic map, a single quadratic recurrence with one tunable parameter, and one of the simplest systems capable of chaos. As the parameter increases, the map passes from a stable fixed point through a cascade of period doublings into chaotic behaviour, and it does so with a structure that is universal across a wide class of unrelated systems. The work follows that route quantitatively: deriving the fixed points and their stability condition analytically and confirming both numerically, extracting the Feigenbaum constant from the cascade, measuring the rate of separation of nearby orbits, and characterising the long-run statistics of a chaotic orbit both in closed form and by coarse-graining the map into a Markov chain, whose stationary distribution is observed to approach that closed form as the partition is refined. The coarse-graining is carried out twice, once with the transition matrix built exactly from the map and once with it estimated from a single trajectory, and the two constructions behave in opposite ways under refinement. That contrast, rather than any individual agreement, is the main finding reported here.

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1 Introduction

1.1 The map

The logistic map is the recursion

$$x_{n+1} = f_r(x_n), \quad f_r(x) = r x(1 - x), \quad (1)$$

with a single real parameter r . It was introduced as a population model, in which x_n is a population expressed as a fraction of a carrying capacity, r is a growth rate, and the factor $(1 - x_n)$ suppresses growth as the population saturates. Its interest is that a rule involving one multiplication and one subtraction produces behaviour ranging from convergence to a single steady state, through cycles of period two, four and eight, to motion that is aperiodic and unpredictable in a precise sense [1, 3].

The tension that makes the map worth studying numerically is this. The dynamics are deterministic, so x_0 fixes the entire orbit. Yet in the chaotic regime the orbit is so sensitive to x_0 that no finite-precision computation can track it for more than a few dozen steps, while the long-run statistics of the orbit, meaning the fraction of time it spends in each part of the interval, are stable, reproducible and available in closed form. The natural objects of study are therefore not individual trajectories but the invariant measure describing where trajectories spend their time, and the Lyapunov exponent quantifying how fast nearby trajectories separate.

1.2 Structure and method

The organising principle is that wherever an exact result is available it is derived first and then used as a test, and only afterwards are the quantities computed that have no closed form.

Sections 2 and 3 are analytical: the parameter range preserving the unit interval, the location of the fixed points, and their stability via the linearisation, with a randomised numerical confirmation of both. Section 4 covers orbits, cobwebs and sensitive dependence. Section 5 presents the bifurcation diagram. Section 6 extracts the Feigenbaum constant by locating superstable parameters with a bracketed bisection. Section 7 computes the Lyapunov exponent and validates it against $\log 2$ at $r = 4$. Section 8 estimates the invariant density from orbits and compares it with the arcsine density. Section 9 builds the Markov coarse-graining, solves for its stationary distribution by a power iteration written from scratch, and compares two constructions of the transition matrix under refinement.

1.3 Results used without proof

Two results are taken from the literature rather than derived here. Isolating them makes the boundary between what is established and what is assumed unambiguous.

- (C1) **Conjugacy at $r = 4$ to the tent map.** The map $f_4(x) = 4x(1 - x)$ is topologically conjugate to the tent map $T(y) = 1 - |2y - 1|$ via $x = \sin^2(\pi y/2)$ [4, 5]. Two consequences are used: the Lyapunov exponent of f_4 with respect to its natural invariant measure is $\log 2$, and its invariant density is the arcsine density

$$\rho(x) = \frac{1}{\pi\sqrt{x(1-x)}}, \quad 0 < x < 1. \quad (2)$$

Both consequences appear only as reference values, never as inputs to a computation.

- (C2) **The Feigenbaum constant.** Its accepted value is $\delta = 4.669\,201\,609\,102\,99\dots$ [2], used in Section 6 as a comparison for a value computed independently here.

Beyond these, a few standard facts are cited where they are used rather than proved, notably the interleaving of superstable and bifurcation parameters in Section 6 and the convergence theory of Ulam's method in Section 9. Everything else in what follows is derived or measured in this work.

1.4 Notation

f_r^n denotes the n -fold composition and $\log$ is the natural logarithm throughout.

2 The admissible parameter range

Definition 2.1 (Forward-invariant set). A set $A \subseteq \mathbb{R}$ is *forward invariant* under f_r if $f_r(A) \subseteq A$. In words: an orbit starting in A never leaves it.

Proposition 2.2. *The interval $[0, 1]$ is forward invariant under f_r if and only if $0 \leq r \leq 4$.*

Proof. Let $x \in [0, 1]$, so $x(1 - x) \geq 0$. Taking $x = \frac{1}{2}$ gives $f_r(\frac{1}{2}) = r/4$, negative whenever $r < 0$, so $r \geq 0$ is necessary; and for $r \geq 0$ we have $f_r(x) \geq 0$ on $[0, 1]$ automatically, so the lower endpoint imposes nothing further.

For $r \geq 0$ the graph is a downward parabola with roots at $x = 0$ and $x = 1$, so its maximum on $[0, 1]$ sits at the midpoint of the roots:

$$f'_r(x) = r(1 - 2x) = 0 \iff x = \frac{1}{2}, \quad \max_{x \in [0, 1]} f_r(x) = f_r\left(\frac{1}{2}\right) = \frac{r}{4}. \quad (3)$$

Hence $f_r([0, 1]) = [0, r/4]$, contained in $[0, 1]$ exactly when $r \leq 4$. $\square$

Remark 2.3. So $r = 4$ is not an arbitrary stopping point but the exact boundary of confinement, and the maximiser $x = \frac{1}{2}$ is the point that reaches it. That same critical point reappears in Section 6 as the basis of the superstable-parameter method. For $r > 4$ an interval around $x = \frac{1}{2}$ maps above 1 and escapes; the surviving set has Lebesgue measure zero [4]. Everything below assumes $0 \leq r \leq 4$ and $x_0 \in [0, 1]$.

3 Fixed points and linear stability

3.1 Location

Definition 3.1 (Fixed point). x^* is a *fixed point* if $f_r(x^*) = x^*$; its orbit is the constant sequence $x_n = x^*$.

Proposition 3.2. For $r \neq 0$ the fixed points of the map are

$$x_1^* = 0, \quad x_2^* = 1 - \frac{1}{r} = \frac{r-1}{r}. \quad (4)$$

The second lies in $[0, 1]$ if and only if $r \geq 1$, and the two are distinct except at $r = 1$, where they coincide.

Proof. The equation $rx(1-x) = x$ rearranges to

$$rx(1-x) - x = x(r - rx - 1) = 0, \quad (5)$$

so $x = 0$ or $rx = r - 1$, giving $x = 1 - 1/r$. The map $r \mapsto 1 - 1/r$ is increasing on $r > 0$, negative for $0 < r < 1$, zero at $r = 1$, and at most $3/4$ for $r \leq 4$, so it lies in $[0, 1]$ precisely when $r \geq 1$. $\square$

Remark 3.3 (On the quadratic formula). Writing (5) as $rx^2 + (1-r)x = 0$ and applying the quadratic formula gives

$$x = \frac{(r-1) \pm \sqrt{(1-r)^2}}{2r} = \frac{(r-1) \pm |1-r|}{2r}, \quad (6)$$

and it is worth being explicit that $\sqrt{(1-r)^2} = |1-r|$ rather than $1-r$, since the two differ in sign on $r > 1$ and the branches are exchanged. Simplifying with either choice of sign gives the two fixed points found previously.

3.2 The stability criterion

Knowing where the fixed points are says nothing about whether an orbit started nearby settles onto one. In the generic case a single number decides it.

Definition 3.4 (Multiplier). The *multiplier* of a fixed point is $\mu := f'_r(x^*)$. Writing $e_n := x_n - x^*$, a Taylor expansion gives $e_{n+1} = \mu e_n + O(e_n^2)$, so to first order the map scales the displacement from x^* by μ each step. A fixed point is *hyperbolic* if $|\mu| \neq 1$, meaning this scaling is not neutral and therefore settles the local behaviour.

Theorem 3.5 (Linear stability). *Let x^* be a fixed point of a continuously differentiable f with multiplier μ . If $|\mu| < 1$ then x^* is locally asymptotically stable and the convergence is geometric with ratio $|\mu|$. If $|\mu| > 1$ it is unstable. If $|\mu| = 1$ the linearisation is inconclusive and higher-order terms decide.*

Proof of the stable case. Since f' is continuous and $|f'(x^*)| < 1$, pick k with $|\mu| < k < 1$ and then $\eta > 0$ with $|f'(\xi)| \leq k$ whenever $|\xi - x^*| \leq \eta$. For any such x the mean value theorem supplies ξ strictly between x and x^* with

$$|f(x) - x^*| = |f(x) - f(x^*)| = |f'(\xi)| |x - x^*| \leq k |x - x^*|. \quad (7)$$

So f maps the interval $|x - x^*| \leq \eta$ into itself and the estimate can be iterated, giving $|x_n - x^*| \leq k^n |x_0 - x^*| \rightarrow 0$. The unstable case follows from the same argument with the inequality reversed and some $k > 1$, for as long as the orbit stays in the neighbourhood. $\square$

3.3 The two multipliers

Differentiating (1),

$$f'_r(x) = r - 2rx = r(1 - 2x). \quad (8)$$

Proposition 3.6. *At $x_1^* = 0$ the multiplier is $\mu_1 = r$, so this fixed point is stable for $0 \leq r < 1$, unstable for $r > 1$, and non-hyperbolic at $r = 1$.*

Proposition 3.7. *At $x_2^* = 1 - 1/r$,*

$$\mu_2 = f'_r\left(1 - \frac{1}{r}\right) = r - 2r\left(1 - \frac{1}{r}\right) = r - 2r + 2 = 2 - r, \quad (9)$$

and hence

$$|\mu_2| = |2 - r| < 1 \iff -1 < 2 - r < 1 \iff 1 < r < 3. \quad (10)$$

So x_2^* is stable for $1 < r < 3$, unstable for $3 < r \leq 4$, and non-hyperbolic at $r = 1$ and $r = 3$.

Remark 3.8 (Sign of the multiplier). The stability range splits at $r = 2$, where $\mu_2 = 0$. For $1 < r < 2$ the multiplier is positive, the error keeps its sign, and orbits approach monotonically from one side. For $2 < r < 3$ it is negative, the error changes sign every step, and orbits approach in an alternating fashion, overshooting each time. The alternating approach is visible directly at $r = 2.8$ in Figure 1, and a one-sided approach in the cobweb at $r = 2$ in Figure 2.

Remark 3.9 (Superstability at $r = 2$). At $r = 2$ the fixed point is $x_2^* = \frac{1}{2}$, which coincides with the critical point of (3). The multiplier vanishes, the linear term of the error recursion disappears, and since $f''_r \equiv -2r$,

$$e_{n+1} = \frac{1}{2}f''_r(x^*)e_n^2 + O(e_n^3) = -re_n^2 + O(e_n^3), \quad (11)$$

so convergence is quadratic rather than geometric. Parameters at which the critical point belongs to the periodic orbit are called *superstable*, and they are the basis of Section 6.

3.4 Numerical confirmation over a random sample of parameters

Both (4) and (9) were derived by hand and are then checked numerically, not at a few hand-picked parameters but across a random sample of the stable range. One hundred values of r are drawn uniformly from $[1.2, 2.8]$, each with an independent random x_0 , and for each an orbit of 1000 iterates is generated. The attained fixed point is taken as the mean of the last fifty iterates and compared with $1 - 1/r$, and the multiplier is estimated from the orbit itself and compared with $2 - r$. Both must agree to within 10^{-4} for the parameter to pass. All one hundred pass, with largest deviations of 1.7×10^{-16} for the fixed point and 2.1×10^{-6} for the multiplier.

Estimating the multiplier from the orbit needs one piece of care worth recording. The prediction $e_{n+1} \approx \mu e_n$ is a linearisation, so the ratio e_{n+1}/e_n is only informative in a window: if $|e_n|$ is too large the quadratic term in (11) is not negligible and the ratio is biased, and if it is too small the

errors are dominated by floating-point noise and the ratio is meaningless. The estimator therefore uses only consecutive pairs with $10^{-12} < |e_n| < 10^{-3}$, and takes the median of the surviving ratios rather than the mean, so that any pair which slips through is not able to dominate. The bounds of that window are set by the two failure modes rather than by tuning against the known answer, but they are not tight: ratios near either end can individually be biased by more than the 10^{-4} tolerance, and it is the median that keeps them from affecting the result.

The interval $[1.2, 2.8]$ is used rather than the full $(1, 3)$ because the convergence rate is $|2 - r|$, which approaches one at both ends of the stable range; near $r = 1$ or $r = 3$ an orbit of 1000 iterates has not converged far enough for the error window above to be populated. That restriction is a consequence of the derived multiplier, not an arbitrary convenience.

3.5 Summary

Range of r	$x_1^* = 0$	$x_2^* = 1 - 1/r$	Behaviour
$0 \leq r < 1$	stable, $\mu_1 = r$	outside $[0, 1]$	extinction
$r = 1$	non-hyperbolic, $\mu_1 = 1$	collides with x_1^*	transcritical bifurcation
$1 < r < 2$	unstable	stable, $\mu_2 \in (0, 1)$	monotone approach
$r = 2$	unstable	superstable, $\mu_2 = 0$	quadratic convergence
$2 < r < 3$	unstable	stable, $\mu_2 \in (-1, 0)$	alternating approach
$r = 3$	unstable	non-hyperbolic, $\mu_2 = -1$	period-doubling bifurcation
$3 < r \leq 4$	unstable	unstable	period two and beyond

Table 1: Fixed-point structure of $f_r(x) = rx(1 - x)$ on $[0, 1]$.

A *bifurcation* is a parameter value at which the qualitative structure of the dynamics changes, and both boundary cases in Table 1 are non-hyperbolic, which is why they are the interesting ones. At $r = 1$ the two fixed points collide and exchange stability, an exchange between colliding branches known as a transcritical bifurcation. At $r = 3$ the multiplier passes through -1 , the signature of a period-doubling bifurcation: the fixed point loses stability and a stable orbit of period two takes its place. The consequences of repeating that mechanism are the subject of Sections 5 and 6.

4 Orbits, cobwebs and sensitive dependence

4.1 Implementation

The orbit generator implements (1) directly: given r , x_0 and a count, it returns the iterates in IEEE double precision, optionally discarding a leading transient. The recursion itself holds nothing subtle; the decisions that matter are structural, and recur throughout the project.

- **Transient discard.** Long-run quantities are computed only after the orbit has settled onto the attractor, and the transient length is chosen so that increasing it does not change the result at the precision quoted.
- **Separation of generation from analysis.** The orbit routine returns raw iterates and performs no reduction of its own, so the same generator feeds the cobweb plots, the bifurcation diagram, the Lyapunov estimator, the histograms and the Markov construction, and a defect in one analysis cannot be masked by a compensating choice in the generator.
- **Explicit handling of degenerate values.** Where a logarithm is taken of a quantity that can vanish, the vanishing entries are excluded explicitly and by name rather than being

allowed to propagate as infinities or be quietly replaced. This matters for the Lyapunov estimator of Section 7, since f'_r vanishes at $x = \frac{1}{2}$.

4.2 Orbit behaviour across the parameter range

Figure 1 shows the first fifty iterates from $x_0 = 0.2$ at six parameters, and most of the regimes in Table 1 can be read from it. At $r = 0.8$ the orbit decays to zero, confirming Proposition 3.6. At $r = 2.8$ it converges to a single value while alternating about it, which is the negative multiplier of Remark 3.8, since $\mu_2 = 2 - 2.8 = -0.8$. At $r = 3.2$ and $r = 3.5$ the orbit settles onto cycles of period two and four. At $r = 3.83$ it settles onto a cycle of period three, a periodic window inside the chaotic region rather than a member of the doubling cascade. At $r = 4$ it is aperiodic.

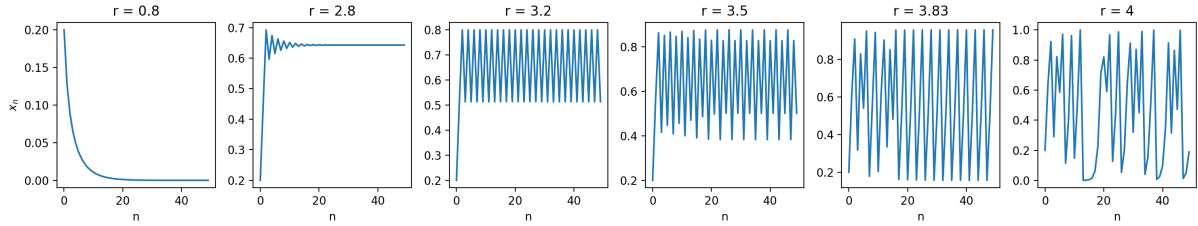


Figure 1: First fifty iterates from $x_0 = 0.2$ at six parameter values, showing extinction, alternating convergence to the fixed point, cycles of period two, four and three, and aperiodic motion.

Figure 2 shows the same behaviour geometrically. A cobweb plot draws the graph of f_r with the diagonal $y = x$ and traces the alternating vertical and horizontal segments representing the recursion, so that fixed points appear as intersections with the diagonal and the local slope at an intersection is the multiplier. At $r = 2$ the orbit reaches the fixed point in very few steps, which is the superstability of Remark 3.9 made visible: the graph is flat where it crosses the diagonal. At $r = 3.2$ the cobweb closes into a rectangle, the signature of a period-two cycle. At $r = 4$ it never closes.

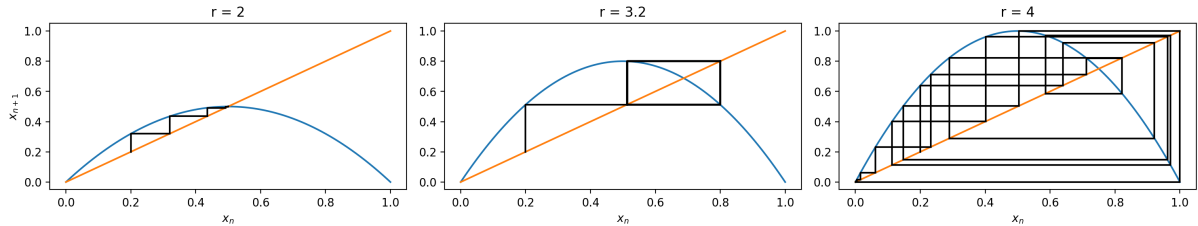


Figure 2: Cobweb plots at $r = 2$, $r = 3.2$ and $r = 4$. The blue curve is f_r , the orange line is $y = x$.

4.3 Sensitive dependence

Two orbits are started from $x_0 = 0.2$ and $x_0 = 0.2 + 10^{-8}$ and their separation is tracked for fifty iterations at two parameters. Figure 3 plots the logarithm of the separation, so an exponential rate appears as a straight line whose gradient is that rate.

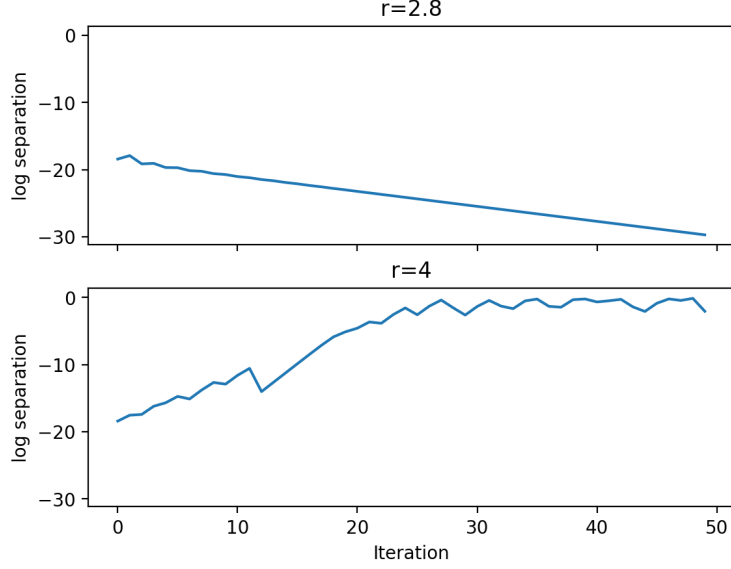


Figure 3: Logarithm of the separation between two orbits started 10^{-8} apart, at $r = 2.8$ (contracting) and $r = 4$ (expanding, then saturating).

Both panels can be checked against results derived above. At $r = 2.8$ the separation contracts, and its decay rate should be $\log |\mu_2| = \log 0.8 = -0.223$ per iteration by Proposition 3.7. The fitted gradient, taken after the first ten iterations once the orbit is near the fixed point, is -0.2232 . At $r = 4$ the separation grows at a fitted rate of 0.681 per iteration over the first 25 iterations, close to $\log 2 = 0.693$, and then saturates at a value of order the diameter of the attractor, beyond which the two orbits are simply unrelated and the separation stops being meaningful. The saturation point is itself predictable: starting from 10^{-8} and doubling each step, saturation is expected after $\log(10^8)/\log 2 \approx 27$ iterations, which is where it occurs.

5 The bifurcation diagram

For each of 1500 values of r spanning $[2.5, 4]$, an orbit of 1500 iterates is generated from $x_0 = 0.4$, the first 1100 are discarded as transient, and the remaining 400 are plotted above that value of r . Where the attractor is a cycle of period p the points collapse onto p values; where it is chaotic they fill out bands.

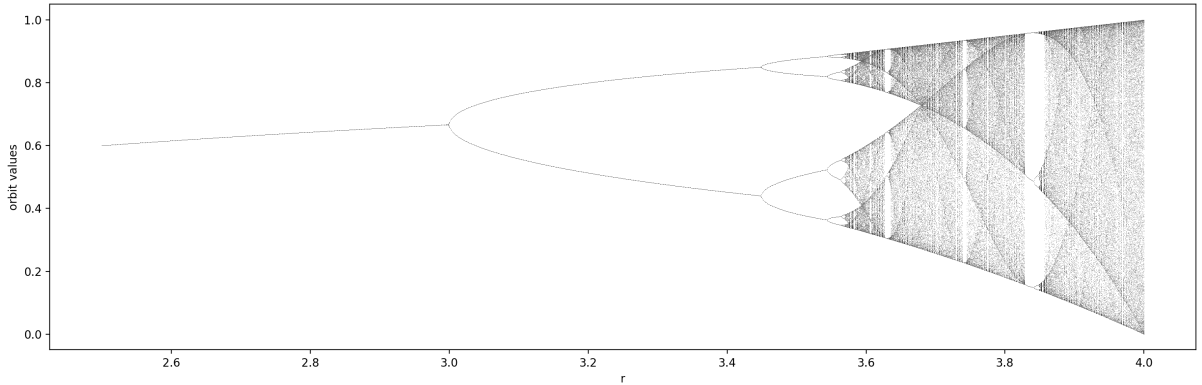


Figure 4: Bifurcation diagram over $r \in [2.5, 4]$. The single branch splits at $r = 3$, the value derived in Proposition 3.7.

Two rendering issues materially affected what the diagram shows and were fixed during the work.

- **Transient contamination.** Too short a transient leaves approach trajectories in the plot, appearing as vertical smearing near the branch points. This is worst precisely where it is least wanted: near a branch point the multiplier approaches 1 in magnitude and convergence is slowest, so the transient requirement is not uniform in r and must be set by the worst case rather than the typical one.
- **Overplotting.** With 1500 parameter values and 400 iterates each, some 600,000 points are drawn, and at ordinary marker sizes the periodic branches saturate and become indistinguishable from the chaotic bands. The marker size and opacity had to be tuned by trial before the structure was resolved. The points are also accumulated into two flat arrays and drawn in a single call rather than plotted per parameter, which is substantially faster.

The first branch point is known from Proposition 3.7: the fixed point loses stability at $r = 3$, and the diagram splits there. This is the last feature that can be checked against a result derived here; beyond it the branch points are roots of polynomials of rapidly increasing degree, and locating them is the numerical problem taken up next.

6 The Feigenbaum constant

6.1 The quantity

The cascade consists of parameters $r_1 = 3 < r_2 < r_3 < \dots$ at which the attracting cycle doubles in period. They accumulate at a finite value and the spacings shrink at an asymptotically constant ratio,

$$\delta = \lim_{n \rightarrow \infty} \frac{r_n - r_{n-1}}{r_{n+1} - r_n}. \quad (12)$$

Feigenbaum's result is that this ratio is identical for a large class of maps with a quadratic critical point, so it is a property of the mechanism rather than of the logistic map [2]. Its accepted value is quoted as (C2); the aim here is to recover it independently.

6.2 Why superstable parameters

The bifurcation parameters r_n are defined by a cycle multiplier equalling -1 . That is numerically awkward: near such a parameter the cycle is only marginally stable, so orbits converge arbitrarily slowly, and locating the cycle in order to evaluate its multiplier becomes ill conditioned exactly where the answer is wanted.

The superstable parameters are a better-conditioned proxy. Define R_n as the parameter at which the critical point $x = \frac{1}{2}$ is itself a member of the cycle of period 2^n , that is the root of

$$g_n(r) := f_r^{2^n}\left(\frac{1}{2}\right) - \frac{1}{2} = 0. \quad (13)$$

Each R_n lies strictly between r_n and r_{n+1} , so the two sequences interleave and accumulate at the same point, and the ratio (12) computed from either has the same limit [2]. The advantages are concrete: g_n is evaluated exactly by 2^n compositions with no iteration to convergence, the cycle is superstable so no stability computation is involved at all, and the defining condition is a sign change in a scalar function, which is the best-conditioned root-finding situation available.

6.3 Method

For each period 2^n up to 256, the parameter R_n is located in two stages.

1. **Bracketing.** A bracket $[a, b]$ with $g_n(a)g_n(b) < 0$ containing no other root is required, and this is the principal failure mode of the method. The function g_n vanishes not only at R_n but at every earlier R_k , since a cycle whose period divides 2^n also returns $\frac{1}{2}$ to itself after 2^n steps, and beyond the accumulation point of the cascade it has many further roots belonging to other superstable cycles. Bracketing the wrong one yields a value that is internally consistent but belongs to a different cycle. The search starts just above the previously located parameter and steps upward to the first sign change; below the accumulation point the next root above R_{n-1} is R_n , so the requirement is that no step passes over it. The step length is the delicate part. A fixed step fails at deeper levels, because the spacing between consecutive superstable parameters shrinks by a factor of roughly δ each time, so a step that is safe at period four will overshoot R_n at period 128, and can overshoot the accumulation point as well, beyond which the extra roots lie. The step is therefore set adaptively to one hundredth of the previous gap, which shrinks at the same rate as the structure being resolved.
2. **Bisection.** The bracket is halved until its width falls below 10^{-12} , converging to the root. Bisection is chosen because, although it converges only linearly, it cannot leave its bracket, whereas a Newton or secant step on a function with many nearby roots can jump to a neighbouring one with no indication that it has done so.

6.4 Results

Successive estimates of the ratio are formed from consecutive triples of located parameters, the first using the superstable parameters of periods one, two and four. Table 2 gives them with their absolute deviation from the quoted value (C2).

Deepest period used	Estimate of δ	Absolute error
4	4.708943	3.97×10^{-2}
8	4.680771	1.16×10^{-2}
16	4.662960	6.24×10^{-3}
32	4.668404	7.98×10^{-4}
64	4.668954	2.48×10^{-4}
128	4.669157	4.44×10^{-5}
256, final estimate	4.669191	1.07×10^{-5}

Table 2: Successive estimates of the Feigenbaum ratio from the located superstable parameters, against the published value $\delta = 4.669\,201\,609\,102\,99$.

The final estimate is 4.669191 against 4.669202, agreeing to five significant figures. The error falls by roughly a factor of four or five per level, which is consistent with the leading correction to (12) itself decaying geometrically in the level index, though the sequence is not smooth enough at these depths to claim that as a measured rate.

6.5 What limits the final estimate

The computation stops at period 256, and it is worth being precise about what limits the result at that depth. The smallest spacing used, $R_8 - R_7$, is about 2.5×10^{-5} , so a bracket width of 10^{-12} fixes each difference entering (12) to a relative precision of order 10^{-7} , well below the 10^{-5} error in the final estimate. That error is therefore dominated by the convergence of the ratio sequence itself rather than by rounding, which is consistent with its continued decrease from level to level in Table 2. Following the cascade further should keep improving the estimate

until the spacings approach the bisection tolerance and the accumulated rounding in evaluating g_n begins to matter. Locating that point empirically, by continuing to longer periods, is the natural extension and has not been done here.

7 The Lyapunov exponent

7.1 Definition and estimator

The Lyapunov exponent measures the average exponential rate at which nearby orbits separate. Linearising along an orbit, a displacement δx_0 evolves to $\delta x_n = \delta x_0 \prod_{k=0}^{n-1} f'_r(x_k)$, so the mean growth rate per step is

$$\lambda = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \log |f'_r(x_n)|, \quad (14)$$

a time average of $\log |f'_r|$ along the orbit. A positive value means exponential separation and hence chaos; a negative value means contraction onto an attracting cycle. The estimator is (14) evaluated over an orbit of 100,000 iterates at $r = 4$ from $x_0 = 0.2$.

7.2 Result

At $r = 4$ the exact value is $\log 2 = 0.6931472$ by (C1). The measured value is

$$\lambda_{\text{measured}} = 0.6931444, \quad (15)$$

a relative difference of 4.0×10^{-6} , or 0.0004%.

This is far closer than a naive error estimate would suggest. Under the arcsine density the terms of (14) have variance $\pi^2/12$, so if they were weakly correlated the statistical error of a 10^5 -term average would be of order 3×10^{-3} , a thousand times the observed residual. The explanation is the conjugacy (C1). Writing $x = h(y)$ with $h(y) = \sin^2(\pi y/2)$, so that $f_4 \circ h = h \circ T$ with $|T'| = 2$, and setting $y_n = h^{-1}(x_n)$, the chain rule gives

$$\log |f'_4(x_n)| = \log 2 + \log |h'(y_{n+1})| - \log |h'(y_n)|, \quad (16)$$

so, up to rounding, the sum in (14) telescopes and the finite- N error is the boundary term $(\log |h'(y_N)| - \log |h'(y_0)|)/N$, of order $1/N = 10^{-5}$. Evaluating this boundary term from the orbit gives -2.7752×10^{-6} , which matches the observed residual $\lambda_{\text{measured}} - \log 2$ to seven significant figures. A consequence is that $r = 4$ is an unusually forgiving test of the estimator: at other chaotic parameters no such cancellation is expected, and the error should fall only as $N^{-1/2}$.

The measurement also confirms, after the fact, the growth rate observed in Figure 3 and the saturation estimate made from it.

8 The invariant density at $r = 4$

8.1 What is being computed

Section 4 established that individual orbits at $r = 4$ are not computable beyond a few dozen steps. The object that is computable, and reproducible, is the distribution of the orbit over the interval. A density ρ is *invariant* if a population of initial points distributed according to ρ remains so distributed after one application of the map. Formally ρ is a fixed point of the transfer operator, also called the Perron-Frobenius operator,

$$(\mathcal{L}\rho)(y) = \sum_{x: f_r(x)=y} \frac{\rho(x)}{|f'_r(x)|}, \quad (17)$$

which pushes a density forward one step: each preimage contributes its own density, weighted by the reciprocal of the local stretching factor, since a region that is stretched has its density diluted. An invariant density satisfies $\mathcal{L}\rho = \rho$.

At $r = 4$ that density is the arcsine density (2), quoted as (C1). It is strongly non-uniform, diverging at both endpoints, so a typical orbit spends disproportionately much of its time near 0 and 1. The divergences are integrable and $\int_0^1 \rho = 1$ as required.

8.2 Estimation

Four independent orbits of 100,000 iterates are generated from four random starting points and each is histogrammed into 200 equal bins. Figure 5 overlays each histogram with (2).

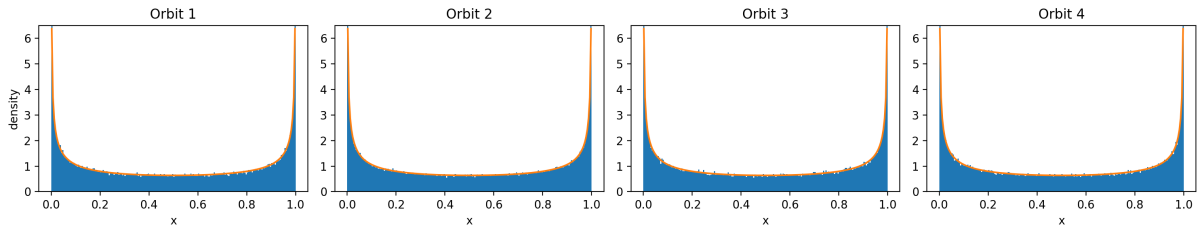


Figure 5: Histograms of four independent orbits at $r = 4$, each of 100,000 iterates from a random start, overlaid with the arcsine density.

Using four different starting points rather than one tests whether the statistics depend on where the orbit starts. The four histograms are visually indistinguishable, and their agreement with the closed form is quantified in Table 3. Strictly, this is evidence of ergodicity, meaning that typical orbits all sample the same measure, rather than a test of invariance itself; the condition $\mathcal{L}\rho = \rho$ is what the discretisation of Section 9 addresses directly. It is also the practical evidence for Section 4 that the statistics do not depend on which particular orbit is followed, despite the individual orbits being pointwise unrelated to one another after a few dozen steps.

Orbit	Mean absolute difference from (2)
1	0.0338
2	0.0254
3	0.0325
4	0.0315

Table 3: Agreement between each histogram and the arcsine density, measured over the central 80% of the interval.

Three points required correction during the work, and each is a general point about comparing an empirical histogram with a continuous density rather than anything specific to this map.

- **Normalisation.** Raw counts must be divided by the number of samples and the bin width, not by the number of samples alone, to be comparable with a density. Dividing by the sample count alone gives a probability mass per bin, which depends on the bin width and does not converge to ρ as the bins are refined.
- **Bin-centre alignment.** The comparison must evaluate ρ at bin centres, obtained by averaging consecutive bin edges, rather than at the edges themselves. A half-bin offset is visually negligible in the flat central region but conspicuous near the endpoints where

ρ varies rapidly, and it does not shrink under refinement as quickly as one might expect, because the error is proportional to the bin width times ρ' , and ρ' diverges.

- **Domain of the error.** The error was originally accumulated over the full interval, including the end bins where ρ diverges. Those bins dominate any norm, and the resulting number measures the singularity rather than the quality of the estimate. All errors reported here therefore use the central 80% of the interval, bins 20 to 180 of 200, and the same convention is carried into Section 9 so that the two are directly comparable.

8.3 What this does and does not establish

The agreement shows that a single orbit, pointwise meaningless after a few dozen steps, nonetheless samples the invariant measure accurately, and that four different orbits sample the same one. It does not constitute a computation of the invariant density from the map: the arcsine form was quoted rather than derived, and the orbits were used only to sample it. Obtaining the density from the map itself, without knowing the answer in advance, requires discretising the operator (17). That is the subject of the next section, and it is where the project stops checking against a known answer and starts producing one.

9 Coarse-graining into a Markov chain: Ulam's method

9.1 The idea

Ulam's method replaces the transfer operator (17), which acts on an infinite-dimensional space of densities, with a finite matrix [6, 7, 8]. Partition $[0, 1]$ into N equal boxes $I_1, \dots, I_N$ and set

$$P_{ij} = \frac{m(I_i \cap f_r^{-1}(I_j))}{m(I_i)}, \quad (18)$$

the fraction of box i that lands in box j . Each row sums to one, so P is a stochastic matrix and the deterministic map has been coarse-grained into a Markov chain on N states: the position within a box is discarded, and what survives is the transition probability between boxes. The discretised invariant density is then the stationary distribution π with $\pi P = \pi$.

This is the conceptual centre of the project. A system containing no randomness has been turned into a probabilistic object, and the information discarded is the resolution within a box. For maps that are uniformly expanding, with $|f'|$ bounded above one, every box is stretched across several boxes at each step, and the stationary distribution is known to converge to the invariant density as the boxes shrink [7, 8]. The map f_4 is not of this kind. Since $|f_4'(x)| = 4|1 - 2x| < 1$ for $|x - \frac{1}{2}| < \frac{1}{8}$, boxes near the critical point are compressed rather than stretched, and the invariant density (2) is unbounded. The convergence theorem therefore does not apply directly here, and whether the coarse-grained description approaches the right answer under refinement is tested empirically in Section 9.5.

9.2 Building the matrix exactly from the map

At $r = 4$ the preimage of a point has two closed-form branches,

$$f_4^{-1}(y) = \left\{ \frac{1 - \sqrt{1 - y}}{2}, \frac{1 + \sqrt{1 - y}}{2} \right\}, \quad (19)$$

so the preimage of a box I_j is a union of two intervals obtained by applying (19) to its endpoints. Note that the left branch is increasing in y and the right branch decreasing, as they must be since f_4 increases on $[0, \frac{1}{2}]$ and decreases on $[\frac{1}{2}, 1]$, so the right-hand preimage interval has its

endpoints in reverse order and must be assembled accordingly. For each box I_i the overlap with each of those two intervals is computed directly, and dividing by the box width $1/N$ converts a length into a probability. This evaluates (18) exactly, by interval arithmetic, with no orbit and no sampling anywhere in the construction.

The stochasticity of P is not imposed but is a consequence of the two branches covering the whole interval, so checking it is a real test of the construction rather than a formality. The largest deviation of a row sum from one is 2.1×10^{-14} , consistent with accumulated rounding alone.

The importance of building the matrix this way is that the resulting stationary distribution is a discretisation of the operator, so its agreement with the arcsine density is independent confirmation, rather than a restatement of the histogram of Section 8.

9.3 Power iteration

The stationary distribution is computed by a power iteration written directly rather than by calling a library eigensolver, since the numerical method is the point of the exercise. From a probability vector v_0 ,

$$v_{k+1} = \frac{v_k P}{\|v_k P\|_1}, \quad (20)$$

iterated until the ℓ^1 change between successive vectors falls below 10^{-10} , subject to a cap of 2000 iterations after which failure is reported rather than a partially converged vector returned. Renormalising at each step is not strictly necessary for a stochastic matrix, whose action preserves the total mass, but it prevents rounding drift from accumulating over hundreds of iterations.

Convergence of this iteration to the dominant left eigenvector, and the uniqueness of the stationary distribution, are properties of the chain rather than assumptions, and are worth testing rather than taking on trust. The iteration was therefore run from two deliberately dissimilar starting vectors, the uniform distribution and a random draw from a Dirichlet distribution, and the two limits agree to within 10^{-8} in ℓ^1 . Figure 6 plots the logarithm of the successive change for both. The two traces are close to straight lines of the same gradient, which is the expected behaviour: the convergence rate of a power iteration is set by the ratio of the second largest eigenvalue modulus to the first, a property of the matrix and not of where the iteration started. Both reach the tolerance within forty iterations.

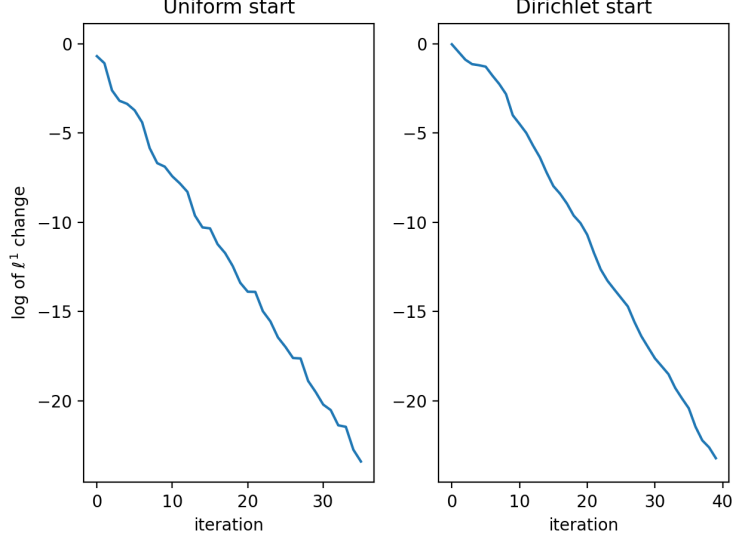


Figure 6: Logarithm of the ℓ^1 change per power iteration, from a uniform start (left) and a random Dirichlet start (right), with $N = 200$.

9.4 The second construction, and why it is kept

The same method can be applied with the matrix estimated from data instead of derived from the map: run a single orbit of $M = 100,000$ points, count the transitions between boxes, and normalise the rows. This is the more obvious construction, and it is the one implemented first.

It contains no new information. Its stationary distribution reproduces the orbit's own occupancy histogram to a maximum absolute difference of 9.7×10^{-6} , which is the diagnostic that identifies the problem: what is being recovered is the Section 8 histogram, arrived at by a longer route. This is an algebraic identity rather than a coincidence: for transitions counted along a single trajectory, the visit frequencies are exactly stationary for the row-normalised count matrix, apart from an end effect from the first and last points of the orbit, which is of order $1/M$. The matrix has N^2 entries estimated from a fixed number of observed transitions, so refining the partition does not refine the estimate, it thins the data.

Figure 7 shows both stationary densities against the closed form at $N = 200$. Both track it well; the map-based one has visible steps near the left endpoint, where the box width is too coarse to resolve a diverging density. The mean absolute difference between the map-based stationary density and the closed form, over the same central 80% used in Section 8, is 0.0331, which is comparable to the histogram errors in Table 3 despite having been obtained without generating a single orbit.

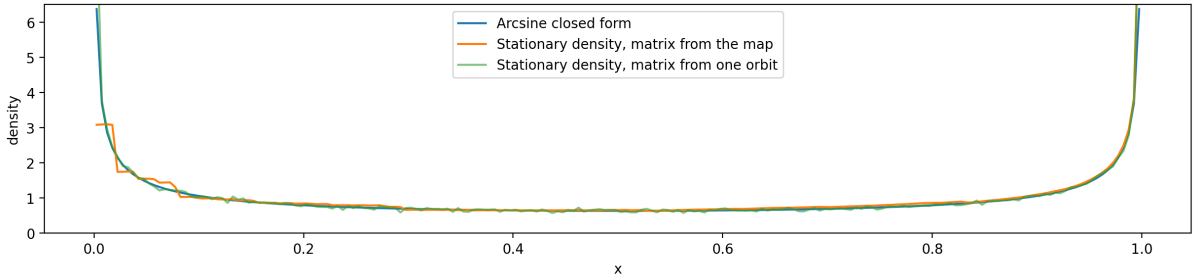


Figure 7: Arcsine closed form against the two stationary densities at $N = 200$, one from the exactly constructed matrix and one from the matrix counted off a single orbit.

9.5 Behaviour under refinement

For box counts from $N = 50$ to $N = 1600$, spaced geometrically at roughly $\sqrt{2}$ per step, the whole pipeline is run at each N : build the matrix, power-iterate to the stationary distribution, convert it to a density by multiplying by N , and compare against the arcsine formula over the central 80% of the interval. The error is the mean absolute difference, and both axes are logarithmic so that a power law appears as a straight line whose gradient is the exponent.

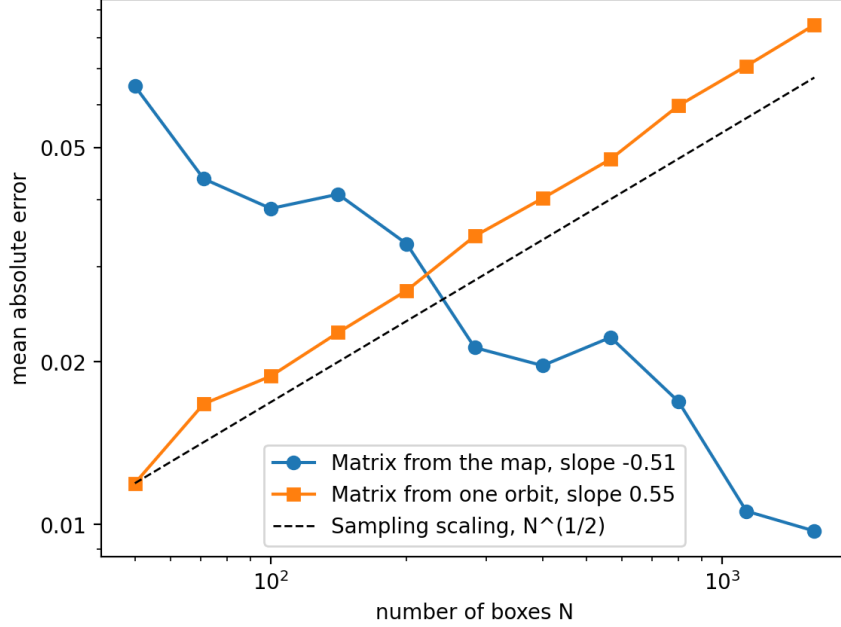


Figure 8: Error against the arcsine density as a function of box count, for the matrix built exactly from the map (circles) and estimated from a single orbit of 100,000 points (squares), with the $N^{1/2}$ scaling expected of sampling error shown dashed, drawn through the first orbit-based point.

Construction of P	Fitted gradient	Interpretation
Exact, from the map	-0.51	error falls as boxes shrink
Estimated, from one orbit	+0.55	error rises as data per box thins

Table 4: Gradient of $\log(\text{error})$ against $\log N$ over $N = 50$ to 1600, for the two constructions.

The map-based curve decreases at a trend of roughly $N^{-1/2}$ over this range. The trend is not perfectly clean: two of the refinement steps tick upward before the decrease resumes, so this is best described as an approximate rate rather than as a measured convergence order, and therefore no claim is made about the asymptotic order.

The orbit-based curve increases, at almost exactly the mirrored gradient. The dashed reference has the $\sqrt{N/M}$ scaling expected of sampling error when M fixed observations are spread across N boxes, and the measured points follow its gradient closely. That agreement is what identifies the cause as sampling noise rather than a defect in the method, and it is why the two constructions were worth running side by side: without the reference line, a rising error curve looks like a broken implementation. In statistical terms this is the familiar trade-off of a histogram built from a fixed sample: narrower bins reduce the bias in each bin but increase its variance, and over this range of N the variance dominates.

The two curves answer different questions. Does the method converge, and does this estimate of it converge? The answers are opposite, and the distinction between them is the point. Refining a discretisation helps only if what is being refined is the operator; refining an estimator built from a fixed quantity of data makes it worse, and the two are easy to conflate when both are described as Ulam's method.

10 Conclusion

Analytically, this project derives the parameter range $0 \leq r \leq 4$ for which the unit interval is preserved, locates both fixed points, derives their multipliers and hence the exact stability ranges $0 \leq r < 1$ and $1 < r < 3$, and identifies the transcritical and period-doubling bifurcations at $r = 1$ and $r = 3$. Both the fixed-point location and the multiplier are then confirmed numerically across one hundred randomly drawn parameters.

Numerically, it produces the bifurcation diagram with the derived branch point $r = 3$ reproduced; recovers the Feigenbaum constant as 4.669191 against the published 4.669202, agreeing to five significant figures, by locating superstable parameters with an adaptively bracketed bisection, with the remaining error set by the convergence of the ratio sequence rather than by rounding; computes the Lyapunov exponent at $r = 4$ as 0.6931444 against the exact $\log 2 = 0.6931472$, and explains the unusually small residual through the conjugacy to the tent map; and estimates the invariant density from four independent orbits, matching the arcsine closed form once the normalisation, alignment and error-domain issues are corrected.

Finally it coarse-grains the map into a Markov chain by Ulam's method, building the transition matrix exactly from the two preimage branches at $r = 4$, solving for the stationary distribution by a power iteration written from scratch and verified to be independent of its starting vector, and measuring the behaviour of the error under refinement of the partition. Built from the map, the error falls at a trend of about $N^{-1/2}$; built from a single trajectory, it rises with the $\sqrt{N/M}$ scaling of sampling error.

A Summary of validation

Computation	Reference quantity	Source of reference
Attained fixed point, 100 random r	$1 - 1/r$	Derived, Prop. 3.2
Measured multiplier, 100 random r	$2 - r$	Derived, Prop. 3.7
Orbit decay rate at $r = 2.8$	$\log 0.8 = -0.223$	Derived, Prop. 3.7
First branch point	$r = 3$	Derived, Prop. 3.7
Cobweb behaviour at $r = 2$	Superstability, $\mu_2 = 0$	Derived, Remark 3.9
Feigenbaum ratio	4.669 201 609...	Cited, (C2)
Lyapunov exponent at $r = 4$	$\log 2$	Cited, (C1)
Lyapunov residual at $r = 4$	Telescoping boundary term	Derived from (C1), Eq. (16)
Orbit separation saturation	$\log(10^8)/\log 2 \approx 27$ steps	Derived from (C1)
Invariant density, four orbits	Arcsine density	Cited, (C1)
Ulam row sums	Exactly 1	Stochasticity of P
Power iteration limit	Independent of start vector	Uniqueness of π
Stationary density, exact Ulam matrix	Arcsine density	Cited, (C1)
Orbit-based stationary vector	Orbit occupancy histogram	Identity, Sec. 9
Orbit-based error growth	$\sqrt{N/M}$	Sampling argument

Table 5: Every result in this document and the independent quantity against which it is checked.

B Computational environment

All computations were performed in Python 3 with NumPy in IEEE double precision, and figures produced in Matplotlib. The companion notebook is `chaos_prob.ipynb`. Running it from top to bottom regenerates every number and figure in this document; its random draws are seeded.

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